

SECURITY AWARENESS PHISHING SIMULATION

ApexPlanet Cybersecurity Internship – Capstone Project

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Project Title

Security Awareness Phishing Simulation Using Kali Linux and Metasploitable 2

Abstract

Phishing attacks remain one of the most effective cyberattack techniques used by threat actors to deceive users into revealing sensitive information or performing actions that compromise security. Organizations worldwide face significant financial and reputational losses due to successful phishing campaigns. Therefore, user awareness and training have become essential components of modern cybersecurity strategies.

This project focuses on developing and demonstrating a Security Awareness Phishing Simulation within a controlled virtual environment. The objective is not to steal credentials or compromise systems, but rather to educate users, monitor interactions, collect evidence, and demonstrate incident response procedures.

The simulation was conducted using Kali Linux as the awareness server and Metasploitable 2 as the user system. Apache2 was used to host a phishing awareness webpage, while Tcpdump and Wireshark were used to monitor and analyze network traffic. Access logs generated by Apache were analyzed to identify user interactions. The project demonstrates how awareness campaigns can be safely conducted while generating valuable security monitoring data.

1. Introduction

Phishing is a social engineering attack that attempts to trick users into performing actions that may compromise the confidentiality, integrity, or availability of information systems. Attackers often disguise themselves as trusted entities and use emails, websites, or messages to manipulate victims.

Despite advances in technical security controls, human error continues to be one of the leading causes of security incidents. Employees may click suspicious links, download malicious attachments, or disclose credentials due to lack of awareness.

Security awareness training plays a crucial role in reducing organizational risk. Simulated phishing campaigns allow organizations to assess user behaviour and improve awareness without causing harm.

This project demonstrates a controlled phishing-awareness exercise where a user accesses an awareness webpage and the interaction is monitored and analysed.

2. Problem Statement

Organizations face increasing threats from phishing attacks, which often exploit human psychology rather than technical vulnerabilities. Many users are unable to identify phishing attempts, resulting in compromised accounts, data breaches, and financial losses.

The objective of this project is to create a controlled phishing awareness simulation that:

- Demonstrates user interaction with awareness content.
 - Collects evidence of user activity.
 - Captures network traffic.
 - Performs security monitoring.
 - Demonstrates incident response procedures.
 - Provides recommendations for improving security awareness.
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3. Project Objectives

The primary objectives of this project are:

1. Deploy an awareness training webpage.
 2. Simulate user interaction in a controlled environment.
 3. Monitor user access through web server logs.
 4. Capture network traffic during the simulation.
 5. Analyze captured packets using Wireshark.
 6. Demonstrate incident detection and response.
 7. Recommend mitigation strategies against phishing threats.
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4. Scope of the Project

Included

- Security awareness webpage deployment.
- User interaction simulation.
- Access log monitoring.
- Packet capture and analysis.
- Incident response documentation.
- Risk assessment and mitigation recommendations.

Excluded

- Credential harvesting.
 - Malware delivery.
 - Exploitation of systems.
 - Unauthorized access.
 - Real-world phishing attacks.
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5. Tools and Technologies Used

Tool	Purpose
Kali Linux	Awareness server and monitoring system
Metasploitable 2	Simulated user machine
Apache2	Web server hosting awareness page
Tcpdump	Packet capture
Wireshark	Packet analysis
VirtualBox	Virtualized lab environment
Linux Networking Utilities	Connectivity testing

6. Lab Environment

Awareness Server

Operating System: Kali Linux

IP Address:

192.168.56.101

Role:

- Hosting awareness webpage
 - Monitoring logs
 - Capturing traffic
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User System

Operating System:

Metasploitable 2

IP Address:

192.168.56.102

Role:

- Simulated user machine
- Access awareness page

Network Configuration

VirtualBox Host-Only Network

Network Range:

192.168.56.0/24

7. Project Architecture

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+-----+
| Kali Linux |
| 192.168.56.101 |
| Apache Awareness Server |
+-----+-----+
|
| HTTP
|
+-----+-----+
| Metasploitable 2 |
| 192.168.56.102 |
| Simulated User |
+-----+
```

8. Methodology

The project was completed in five phases:

Phase 1 – Planning

The objectives, scope, tools, and network architecture were identified.

Activities:

- Define project goals.
 - Configure virtual environment.
 - Assign IP addresses.
 - Establish communication between systems.
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Phase 2 – Implementation

Apache2 was installed and configured on Kali Linux.

A custom awareness webpage was developed containing:

- Phishing indicators.
- Safe browsing practices.
- Security recommendations.

The webpage was deployed under:

`/var/www/html/index.html`

Phase 3 – User Interaction Simulation

The user system (Metasploitable 2) accessed the awareness webpage hosted on Kali Linux.

Connectivity was verified using:

- Ping
- Wget

Successful access generated web server logs.

Phase 4 – Monitoring and Detection

Monitoring was performed through:

Apache Access Logs

Logs recorded:

- Source IP
- Timestamp
- Requested resource

- HTTP status code

Example:

192.168.56.102 - - "GET / HTTP/1.1" 200

Packet Capture

Tcpdump was used to capture traffic between the two machines.

Captured traffic included:

- HTTP Requests
 - HTTP Responses
 - Source and Destination IPs
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Phase 5 – Analysis

Captured traffic was analyzed using Wireshark.

The analysis confirmed:

- Source IP: 192.168.56.102
- Destination IP: 192.168.56.101
- Protocol: HTTP
- Request Type: GET
- Response: HTTP 200 OK

The analysis verified successful interaction between the user system and awareness server.

9. Evidence Collected

The following evidence was collected during the project:

1. IP configuration screenshots.
2. Connectivity test screenshots.
3. Awareness webpage screenshots.
4. Apache access log screenshots.
5. Tcpdump capture screenshots.
6. Wireshark analysis screenshots.
7. Packet capture (.pcap) file.

These artifacts demonstrate successful completion of the simulation.

10. Incident Response Process

Detection

The Apache web server recorded incoming requests from the user machine.

Detection Sources:

- Access Logs
 - Tcpdump
 - Wireshark
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Analysis

The observed activity was analyzed.

Findings:

- User successfully accessed awareness content.
 - No malicious payloads were delivered.
 - No credentials were collected.
 - Activity occurred within a controlled environment.
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Containment

Actions Taken:

- Traffic monitored.
 - Logs preserved.
 - Evidence collected.
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Eradication

No malicious components were identified.

No eradication procedures were required.

Recovery

Recovery activities included:

- Verification of server availability.
 - Review of collected logs.
 - Archiving of evidence.
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Lessons Learned

- Security awareness is essential for reducing phishing risk.
 - Logging provides valuable forensic information.
 - Network monitoring improves visibility.
 - User education remains a key defense mechanism.
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11. Findings

Finding 1

Awareness webpage successfully deployed and accessible.

Severity: Low

Finding 2

User interaction successfully simulated.

Severity: Informational

Finding 3

Web server logs successfully captured user activity.

Severity: Low

Finding 4

Network traffic successfully captured and analyzed.

Severity: Informational

12. Risk Assessment

Risk	Likelihood	Impact
User clicks phishing link	High	High
Lack of awareness	High	High
Insufficient monitoring	Medium	High
Delayed incident response	Medium	Medium

13. Recommendations

1. Conduct periodic phishing awareness training.
 2. Implement Multi-Factor Authentication.
 3. Deploy email filtering solutions.
 4. Establish incident reporting procedures.
 5. Enable centralized log monitoring.
 6. Deploy Security Information and Event Management (SIEM) systems.
 7. Perform regular awareness assessments.
 8. Conduct periodic security audits.
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14. Conclusion

The Security Awareness Phishing Simulation project successfully demonstrated how phishing-awareness exercises can be conducted within a controlled environment. The project involved webpage deployment, user interaction simulation, log monitoring, packet capture, traffic analysis, and incident response documentation.

The results highlight the importance of user education, continuous monitoring, and evidence collection in defending against phishing attacks. The project achieved all defined objectives and provided practical experience in web server administration, network analysis, security monitoring, and incident response procedures.

This exercise demonstrates that awareness training, combined with technical monitoring controls, can significantly improve an organization's resilience against phishing threats.